
ARI: An Index for Measuring Reproducibility in Deployed ML Systems

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<https://github.com/semq/semq-ari-benchmark>

Abstract

Two identical requests to the same embedding API can return numerically different vectors, yet the frequency and magnitude of this variation are typically not reported by providers. We introduce the AI Reproducibility Index (ARI), a protocol for measuring output reproducibility by deterministically quantizing model output vectors and measuring code equality under predefined execution conditions. Under a fixed protocol, mean exact-code agreement across five commercial embedding APIs ranges from 0.169 to 1.000: one API produces identical codes across repeated calls under process restarts, concurrent load, and time gaps, whereas with another, 83.1% of input-condition comparisons differ. Precision and serving perturbations alter the quantized representation for 47–100% of corpus documents while Recall@10 remains statistically indistinguishable from baseline. Yet small local differences compound across multi-step computation: at a perturbation level yielding Recall@10 = 0.988, 11.5% of 15-hop retrieval trajectories diverge, while 0.48–0.79% per-token divergence under bf16 produces a different complete greedy generation 20.8–35.4% of the time. These results show that reproducibility is a property of a deployed ML system that conventional endpoint quality metrics can fail to resolve, and that small local differences can compound into substantially larger outcome divergence across sequential computation.

1 The measurement gap

Reproducibility in deployed inference depends on execution conditions. Changes in numerical precision, floating-point reduction order, kernel selection, hardware, or batching can alter model outputs, while a nominally fixed inference stack may also vary when its computation is not batch invariant [7]. Sampling randomness is excluded: the variation considered here arises from numerical execution and serving conditions. In our experiments (§3.2, §3.3), such changes range from small floating-point differences to per-component perturbations on the order of 10^{-4} .

Retrieval metrics such as Recall@10 compress a high-dimensional retrieval computation into a coarse task-level statistic, so representations and retrieved lists can change without a statistically distinguishable change in the metric. In §3.2, precision and execution-condition changes alter the quantized representations of 47–100% of documents, and in some conditions alter most top-10 lists, while Recall@10 remains statistically indistinguishable from the reference configuration. For hosted APIs, users additionally lack control over backend precision, batching, kernel selection, and hardware, so they cannot attribute observed drift to a particular serving mechanism.

Prior work documents numerical variation in hosted embeddings [8], precision-dependent embedding reproducibility [1], and batch-dependent inference numerics [7]; we address a different problem — measuring such variation under a common cross-system protocol. ARI applies a bit-exact representation-level comparison to a frozen input set, reports code agreement with bootstrap intervals, and operates without access to a provider’s serving infrastructure, provided the system exposes a vector-valued representation. Governance frameworks similarly motivate reliable, con-

40th Conference on Neural Information Processing Systems (NeurIPS 2026) Workshop: Machine Learning for Systems.

sistent, monitorable AI behavior and the development of measurement methodologies [2, 3, 4], but none we examine specifies a representation-level metric for reproducibility.

2 The ARI protocol

Definitions. For representation-level measurement, the system under test is $A : \mathcal{X} \rightarrow \mathbb{R}^d$, mapping an input to an embedding or per-step logit vector. A baseline run holds the specified caller-controlled execution axes fixed, and each comparison condition varies one such axis. On self-hosted systems, this can approximate a single-variable intervention. For hosted APIs, backend state is unobservable; drift is therefore reported as associated with the experimental condition rather than attributed to a specific backend mechanism.

id	varied condition	variation potentially exposed
same	immediate repeat	within-session repeatability
proc	fresh process	process-dependent execution (self-hosted); client state (API)
conc	concurrent burst	load-associated batching, composition, routing, queueing
time	wall-clock gap > 24 h	time-associated drift; backend cause unobserved
prec	fp32 $\rightarrow$ bf16/fp16	precision-induced drift (self-hosted diagnostic)
mach, lib	host; library version	hardware/library drift (self-hosted diagnostics)

The probe. Let Q be a deterministic per-coordinate quantizer and $P = Q \circ A$. We use a four-level symmetric scalar quantizer (SEMQ QBIN) with one model-specific scale s , calibrated once as the 99th percentile of absolute coordinate values pooled over a published reference batch. Each coordinate is represented by two semantically defined binary operators, $Q_s(x_j) = (\mathbf{1}[x_j \geq 0], \mathbf{1}[|x_j| \geq s/2])$, corresponding to sign and magnitude region, for exactly two bits per dimension. The interior edge falls at $s/2$ because two equal-width magnitude bins partition the calibrated range $[0, s]$ (p99 gives a corpus-stable s). The probe is deterministic, so the instrument introduces no stochastic disagreement; its expected response is approximately linear for sufficiently small injected perturbations; and its sensitivity is coordinate-local, so perturbing one input coordinate cannot alter unrelated code coordinates (Appendix D).

For calibration, we perturb representations as $x' = x + \epsilon$ with $\epsilon \sim \mathcal{N}(0, (\sigma \overline{\|x\|_2})^2 I_d)$, where $\overline{\|x\|_2}$ is the mean representation norm over the reference batch and σ is the per-coordinate noise standard deviation expressed as a fraction of that norm (dimensionless). Noise is added before any embedding re-normalization. Let ρ denote the coordinate change fraction: the fraction of coordinates whose two-bit code differs. At small perturbations a changed coordinate typically flips one of its two indicators (sign near zero, magnitude near $\pm s/2$), so $\rho \approx \bar{H}/d$ for $\bar{H}$ the mean changed-bit count ($\bar{H}/2d \leq \rho \leq \bar{H}/d$ exactly). Within the calibrated regime,

$$\mathbb{E}[\rho] = \kappa \sqrt{2d/\pi} \sigma^b,$$

where b and κ are model-specific and fitted per model. Across 13 architectures the fitted exponents have mean 0.979 and standard deviation 0.028, while $\kappa \in [0.37, 0.76]$.

Metrics. ARI names the protocol; it reports two complementary quantities. ARI-HER is mean exact-code agreement over $\{\text{proc}, \text{conc}, \text{time}\}$, the externally inducible core: strict whole-code repeatability, where HER is the fraction of inputs whose complete quantized codes agree exactly. ARI- $\hat{\sigma}$ is the calibrated noise-equivalent magnitude defined below. Exact whole-code agreement depends on dimension and model-specific probe sensitivity, so raw HER is not by itself a normalized cross-model measure; for responses within the calibrated regime we therefore also report

$$\hat{\sigma} = \left(\frac{\rho}{\kappa \sqrt{2d/\pi}} \right)^{1/b},$$

a per-coordinate noise-equivalent scale relative to mean representation norm. It normalizes the probe-response model’s dependence on dimension and sensitivity — not the vector-level perturbation, which at fixed σ grows as $\sqrt{d}$ — and it is not a direct estimate of arbitrary serving noise (§3.1, Appendix G). Treatment semantics differ where provider internals are inaccessible, and the core set does not cover unobserved backend configuration changes.

Replication. Each condition is one execution pair over the 1,000 inputs. Intervals are a nonparametric bootstrap over inputs, resampling whole input rows (repeated passes of an input stay together);

they quantify input-set uncertainty *conditional on that pair* and do not capture environmental variance across independent days, bursts, or routing realizations. A single 76-hour comparison can detect time-associated drift but cannot estimate its temporal frequency or variance.

Layers. The vector-valued definition covers ARI-R (embeddings) and ARI-D, which scores the full-vocabulary logit vector at each decoding step under teacher forcing — every condition is fed the baseline greedy token sequence, so per-step logits respond to identical prefixes; free-running generation is reported separately (§3.3). ARI-E extends the design to discrete end-to-end outcomes and measures verdict agreement directly, without quantization.

Verifiability. Inputs are a frozen, hash-pinned 1,000-item BEIR slice spanning medical, scientific and financial domains. A report carries per-condition HER, $\bar{H}$ and CI, an environment manifest, and a SHA-256 digest over the code matrix in fixed input order; an Ed25519 signature covers a canonical JSON manifest carrying the report’s SHA-256 and the input digest, and the standalone verifier has no third-party dependencies and does not import the ARI implementation. A report is thus tamper-evident — a registered signer attested to this result over this input set — which is weaker than independently reproducible: a third party cannot reconstruct a provider’s backend to re-run a historical API call.

3 Results

3.1 Reproducibility is provider-specific and spans the full range

Five commercial APIs, one protocol, frozen inputs, $n = 1,000$; time at a ~ 76 h gap, conc a 64-way burst vs calm:

provider / model	same	proc	conc	time	ARI-HER	95% CI
gemini-embedding-001	1.000	1.000	1.000	1.000	1.000	[1.000, 1.000]
text-embedding-3-large	0.859	0.862	0.851	0.822	0.845	[0.822, 0.866]
mistral-embed	0.734	0.753	0.791	0.554	0.699	[0.672, 0.727]
voyage-4-large	0.571	0.615	0.209	0.676	0.500	[0.472, 0.528]
embed-v4.0	0.133	0.146	0.130	0.232	0.169	[0.147, 0.194]

The spread is the result. One provider is bit-reproducible and another drifts on more than 80% of repeated calls under identical inputs, and a caller cannot infer which from anything providers publish today.

Dimension and probe sensitivity partly shape these numbers. Recomputed as the noise-equivalent $\hat{\sigma}$ of \$2 (the five models span $d \in \{1024, 1536, 3072\}$, $\kappa \in [0.37, 0.66]$), the top three hold their order while the two lowest reorder, landing within about 30% of each other (Appendix G). The three-fold HER gap between them is largely the exact-match penalty on a higher-dimensional code; class conclusions stand, and comparison between low-HER providers should read $\hat{\sigma}$.

Two checks support it. The 1.000 is not a cache: 150 novel nonce-bearing inputs held that provider at 1.000 while a control provider read 0.807. And single treatments carry distinct failures: only conc exposes one provider collapsing $0.615 \rightarrow 0.209$ under burst, and only a multi-day time gap exposes another falling to 0.554, invisible at 19 hours. Float-level residuals: Appendix F.

Eight self-hosted open models read 1.000 across proc, conc, and time, including with BLAS threads unpinned, and every one collapses to $\text{HER} \leq 0.094$ under a bf16 serving change. Across the tested self-hosted models, pinning precision was sufficient to eliminate the measured core-condition drift.

3.2 Recall@10 can remain unchanged under large representation drift

BEIR SciFact (5,183 documents, 300 judged queries), all-MiniLM-L6-v2; reference fp32/CPU/4-thread/batch-32, every other condition changing one thing in a fresh process. Intervals: paired bootstrap over queries, 10,000 resamples.

Neither precision change moves Recall@10 detectably. int8 moves mean cosine similarity against the fp32 reference to 0.938, and its 0.6-point recall drop sits well inside the noise band, while HER reads a hard zero: no document among 5,183 keeps its code. The control rows make that reading

condition	$\Delta R@10$	95% CI	sig.	top-10 same	HER
new proc / 1 thread / batch 8 / batch 128	+0.0000	[+0.0000, +0.0000]	no	100.00%	1.0000
GPU fp32, TF32 off	+0.0000	[+0.0000, +0.0000]	no	100.00%	0.9992
GPU fp32, TF32 on	+0.0000	[+0.0000, +0.0000]	no	96.33%	0.5267
bf16	+0.0033	[+0.0000, +0.0100]	no	45.00%	0.0000
int8	-0.0061	[-0.0278, +0.0156]	no	0.00%	0.0000

meaningful: process, thread, and batch changes move nothing on any panel, so HER = 0 under a precision change is a discrimination, not an instrument that fires on everything.

The TF32 pair is the operationally important row. A model audited at fp32 on CPU and served at fp32 on GPU gets a different code for 47% of documents, with no configuration change visible to the caller. Turning TF32 off restores 0.9992, which leaves one switch; a 2x2 isolation over TF32 $\times$ use_deterministic_algorithms confirms it (cross-process HER 1.000 off, 0.749 on; the determinism flag has no effect). TF32’s 10-bit mantissa turns a $\sim 10^{-7}$ cross-process kernel difference into $\sim 10^{-3}$, above the quantizer floor.

One scope note: the aggregate metric is blind, result lists are not — int8 changes every top-10 list and bf16 55%, so diffing stored lists detects these events, at the cost of a stored reference top- k , saved-query-only coverage, and tie-order churn.

3.3 Small per-step divergence compounds into outcome divergence

Iterative retrieval (controlled stress test). A self-avoiding nearest-neighbor walk over NFCorpus (3,633 documents) embeds the query at each hop and steps to the nearest unvisited document — an LLM-free iterative-RAG model. We inject *synthetic* isotropic Gaussian per-hop noise over 15-hop walks (100 seeds $\times$ 30 runs), with $\sigma = 10^{-3}$ the largest value still passing the standard check (recall@10 = 0.988 $\geq$ 0.98); 11.5% of trajectories diverge by depth 15 versus 0.002 at depth 1 — an 8.6x amplification of the per-hop top-1 flip rate; at $\sigma = 0$ divergence is 0 everywhere. This Gaussian need not match real API residuals’ covariance, anisotropy, or temporal correlation: it bounds what per-hop drift *can* do; replaying empirical residuals is next.

Decoding. The same protocol one layer up, on the full-vocabulary logit vector at each greedy step (48 prompts, $\sim 2,265$ scored steps each, fp32 reference, same GPU):

model	condition	token agreement	HER	generations identical
Llama-3.1-8B	TF32 off (control)	100.00%	1.0000	48/48
Llama-3.1-8B	TF32 on	100.00%	0.0000	48/48
Llama-3.1-8B / Qwen2.5-7B / Mistral-7B	bf16	99.21–99.52%	0.0000	38 / 31 / 38 of 48

A serving change can be total at the instrument and invisible at the output: under TF32 the code changes at every step while all 48 generations stay character-identical. Small per-token loss is large per-outcome loss: 0.48–0.79% of tokens flipped under bf16 becomes 20.8–35.4% of generations changed — a 26–45x amplification; one flipped token derails everything after it. The caveat is the reverse: over a $\sim 100k$ -token vocabulary, whole-code equality saturates HER at zero on irrelevant differences, so $\bar{H}$ and $\hat{\sigma}$ carry the comparison.

3.4 At the outcome layer, the harness is part of the system

ARI-E applies the same discipline to agent outcomes on `nvidia/Open-SWE-Traces`: two scaffolds $\times$ two models, $\sim 18,000$ SWE-bench-style instances graded by each repository’s tests. The estimator is an agreement gap: on instances with ≥ 2 rollouts per side, $\Delta = D_{\text{cross}} - D_{\text{self}}$, the paired-verdict disagreement rate across conditions minus the same rate within one. Under Qwen3.5-122B, $\Delta = 0.215 - 0.126 = +0.089$ [+0.083, +0.094]; across four contrasts $D_{\text{self}}/D_{\text{cross}}$ is 59–74%, and the mean scaffold effect (+0.069) exceeds the mean model effect (+0.050). Δ is descriptive, not causal (assumptions: Appendix E).

Limitations.

Decoding runs are greedy (drift-vs-sampling separation above temperature 0 untested) and decoding conditions are serving changes applied on one machine, not observed provider changes; Appendix B lists the rest.

4 Conclusion

Rank-based endpoint metrics cannot measure reproducibility; a deterministic quantizer over the output representation can: five commercial APIs span 0.169 to 1.000, a precision change rewrites half to all of a corpus’s codes without moving Recall@10 detectably, and per-step drift too small to see becomes 11.5% of trajectories and 35% of generations. The compounding is why the first two numbers matter.

References

- [1] ReproRAG: reproducibility of retrieval-augmented generation embeddings. arXiv:2509.18869, 2025.
- [2] Regulation (EU) 2024/1689 (Artificial Intelligence Act), Article 15: accuracy, robustness and cybersecurity.
- [3] NIST. Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1, January 2023. (Voluntary framework; Manage 4.1 concerns post-deployment monitoring.)
- [4] US Treasury AI Risk Management Framework, February 2026.
- [5] Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., and Gurevych, I. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. NeurIPS Datasets and Benchmarks Track, 2021.
- [6] Muennighoff, N., Tazi, N., Magne, L., and Reimers, N. MTEB: Massive Text Embedding Benchmark. EACL, 2023.
- [7] Thinking Machines Lab. Defeating Nondeterminism in LLM Inference, 2025. Argues that many inference kernels are run-to-run deterministic but not batch-invariant, so batch composition, not atomics alone, drives much observed LLM nondeterminism.
- [8] openai/openai-python, issue #868: community-reported nondeterministic embeddings, 2023—.
- [9] NVIDIA. Open-SWE-Traces dataset (Hugging Face).
- [10] The SEMQ Group. ARI-Bench-v0.1 and the ARI reference implementation. Content hash e9ec8b01c62635de.

A What our own testing took away

Three earlier claims did not survive adversarial testing, and the corrections bound what the index needs. At the decoding layer, an 8-byte top-2 logit margin recovers 86% of the quantizer’s signal for 1/2000th of the stored reference state and is equally bit-reproducible; the quantizer’s remaining advantage there is coverage of drift below the top ranks, demonstrated for token suppression (28.7x enrichment at ranks 3–10, with 100.00% token agreement, so output monitoring is blind to it) and unproven for serving changes generally. The honest framing is a tradeoff, not a defeat: ARI-D is a high-sensitivity diagnostic instrument, and a top-rank margin is the cheaper monitor when only output-relevant perturbations matter. A frozen product-quantizer codebook also works as a probe: on every real k-means codebook we swept, $\text{encode}(\text{reconstruct}(c)) = c$ held at every setting, including one with 91% of assignments within 10^{-6} of a tie. What quantile binning offers is a verification-cost difference — a guarantee checkable from the system owner’s declared form in advance, and one scalar of calibration state instead of thousands of floats — and that is an argument about cost, not an impossibility result. The contribution is the measurement protocol, not a claim that QBIN is the only usable deterministic probe. And ARI-E uses no quantizer at all, since nothing in a verdict comparison is a vector.

B Remaining limitations

mach and lib are unmeasured for the self-hosted panel; one provider’s conc ran at burst 16 rather than 64 due to rate limits; GPU coverage is two architectures. Drift rates are input-distribution dependent, which is exactly why the input set is frozen and hash-pinned. About 22% of ARI-E rows

were ungraded and dropped; if grading failure tracks task difficulty, the surviving set is easier than the whole.

C Decoding controls

The model whose vocabulary fits one probe context behaves like the two that require chunking, so the chunked probe path introduces no artifact.

D Probe sensitivity spread

Measured influence spread is 1.00 for quantile binning versus about 16 for a block-microscaled format like NVFP4, whose shared per-block scale smears one perturbed dimension across its whole block. A diagonal sensitivity Jacobian is what lets drift attribute back to specific dimensions.

E ARI-E estimator assumptions

$\Delta = D_{\text{cross}} - D_{\text{self}}$ is a descriptive agreement gap. It assumes rollouts within a condition are exchangeable, so that D_{self} estimates the disagreement a fixed condition generates against itself; it is defined only on instances with at least two rollouts per side, since one rollout gives no self-consistency control; and it is not a causal estimate of a scaffold’s effect on task success. It answers a narrower question: how much of the disagreement between two conditions exceeds what either condition produces against itself. About 22% of rows were ungraded and dropped (Appendix B).

F Float-level residuals

Of 30 texts embedded in two separated passes against one drifting provider, 3 differed; where they differed, about 2,300 of 3,072 dimensions changed at magnitude $\sim 10^{-4}$. This is consistent with backend-level numerical variation rather than isolated last-bit differences, though without provider telemetry we cannot attribute it to a specific mechanism.

G Normalized panel ordering

Recomputing the API panel as the noise-equivalent perturbation of §2, over the externally observable set {proc, conc, time}, using the per-model (d, b, κ) (coordinate-level convention; the registry stores 4κ). The reports store $\bar{H}$ in bits; we use $\rho \approx \bar{H}/d$, the low-drift coordinate change fraction of §2. $\hat{\sigma}$ is invariant to the unit convention; the residual approximation error from the ρ bounds is model-independent up to the spread in b , a distortion under 20% — smaller than every gap read from the table.

model	d	κ	b	ARI-HER	$\bar{H}$	$\hat{\sigma}$
gemini-embedding-001	3072	0.637	0.990	1.000	0.000	0
text-embedding-3-large	3072	0.413	0.932	0.845	1.363	1.1×10^{-5}
mistral-embed	1024	0.371	0.915	0.699	1.349	6.1×10^{-5}
voyage-4-large	1024	0.622	0.979	0.500	9.566	5.0×10^{-4}
embed-v4.0	1536	0.664	0.995	0.169	12.200	3.7×10^{-4}

The top three hold their order with clear separation. The two lowest reorder: embed-v4.0’s $\hat{\sigma}$ is about 26% below voyage-4-large’s, a gap comparable to the method’s own distortion, so the defensible statement is that the two are of similar drift magnitude — not, as raw HER suggests, three-fold apart. $\hat{\sigma}$ is a per-coordinate noise-equivalent scale under the isotropic-Gaussian calibration model; under the vector-level alternative $\hat{\eta} = \sqrt{d} \hat{\sigma}$ the ordering is unchanged and the two lowest providers

move within 10% of each other. Real serving residuals need not be isotropic, so these values support ordering and rough magnitude, not precise recovery of backend noise.

Reproducibility statement

The specification (probe, calibration rule, condition set, report schema, per-model fingerprints), the frozen 1,000-item input set with its content hash, the reference implementation, the independent verifier, and every experiment's protocol with machine-readable results are in the accompanying repository; each experiment directory carries a reproduce command. Figures are generated from the experiments' own JSON output, so no figure recomputes a number independently of the table it illustrates. Code is Apache-2.0; the specification and input set are CC-BY-4.0, with BEIR text retaining its original licenses.